The Relational Tick: How Event Density Builds Local Clocks Without a Universal One

Mass as carried memory, and V5 as the finite-reach binder of local proper time

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Preamble: what this paper does NOT claim

Written first, per house discipline; the abstract is reconciled against it.

1. It does not derive the value of any mass, coupling, or timescale. All results here are form-level; numerical values are inherited, following the standing Event Density (ED) pattern (form FORCED, value INHERITED).
2. It does not derive, from the 13 primitives, that the substrate must carry the V5 cross-chain coupling. V5 is implemented as an honestly-named structural addition, faithful to its corpus-stated form (finite cross-chain reach, finite memory, retarded support, gauge-covariant relative phase), but the forward derivation of V5 from the primitives remains open. This paper characterizes what V5 does and what structure it must have, not that the primitives force it.
3. It does not claim the chain-carried memory channel is present in the certified reference substrate. The certified base is ballistic-or-extinct, its update driven by commitment density alone, with no chain-specific memory and no cross-chain coupling. The memory channel and the V5 coupling are additions; their necessity is argued from the physics they produce, not derived.
4. It does not claim a Higgs mechanism, electroweak symmetry breaking, gauge-boson mass, or a condensate. The mass result here is an intrinsic inertia (a chain's carried memory slows it), not the Standard Model Higgs sector.
5. It does not fit stellar pulsation, quantitative time dilation, or the quantitative Zeno rate. The connections to those are structural (same mechanism, correct qualitative behavior), not numerical fits.
6. The sparse-becoming spine of Section 2 is not new. It is already derived in the ED gravity sector and is cited here as prior work, not presented as this paper's contribution.

Abstract

Event Density has no universal tick. The primitive most easily mistaken for one, P13, is time homogeneity: a symmetry statement, that the substrate laws are invariant under time translation with no privileged moment, not a clock. Physics runs on commitment (P11), the sparse, local, irreversible becoming events, which cannot fire at every fundamental step without freezing all dynamics in the quantum Zeno limit. This prohibition on universal becoming is load-bearing three ways at once, a result already derived in the ED gravity sector: dense commitment would impose the Zeno freeze (so quantum mechanics forces sparse commitment); gravitational time dilation is becoming at a position-dependent rate, impossible if becoming were uniform; and the preferred-frame parameter that would signal a universal frame is driven far below observational bounds by that same sparseness (a mechanism estimate of order 10^{-93}, not a certified number). The corpus establishes why becoming must be sparse and local. This paper supplies the missing constructive half: how local shared clocks (bound composite objects) form inside a world that forbids a universal one. The mechanism is V5, ED's finite-reach cross-chain kernel. We show, first in a phase model and then in the certified substrate simulator, that V5's finite reach binds nearby chains into a shared rate (a local proper time, the substrate face of a composite rest frame) while leaving distant chains free (time dilation preserved,

no universal clock), and that this behavior hinges on the reach being finite. We further show that mass, in this picture, is a chain's carried memory of its own commitment history, and that a chain persisting by re-committing acquires a real, noise-robust temporal rhythm. Mass and time dilation emerge as one phenomenon viewed from V5: chains acquiring a shared local rate of becoming.

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1. The result: V5's finite reach builds local clocks, not a universal one

ED's gravity sector requires that physical becoming be sparse and local, never universal (Section 2). A world with that constraint faces an immediate question: if there is no universal clock, how do ordinary bound objects, which plainly share a common rest-frame time, come to exist at all? Something must bind local collections of chains into a shared local rate of becoming without ever extending that sharing to a universal scale. We identify that binder as V5, the cross-chain kernel of Paper_090, whose defining feature is a finite cross-chain reach.

V5's known structure. Paper_090 gives V5 as a retarded, cross-chain kernel with a finite reach length (its influence decays with cross-chain separation), a finite memory time, and a gauge-covariant relative phase. The reach being finite is not incidental; it is the load-bearing feature for what follows.

The test. Take chains with genuinely different natural rates of becoming, arranged in two spatially separated groups, the groups themselves running at different mean rates (a time-dilation setup: two composites in relative motion or different potential). Couple them with a finite-reach cross-chain term of V5's form. Ask: does the coupling bind nearby chains into a shared rate (local proper time) while leaving distant groups free (no universal clock)?

Phase-model result. In a minimal phase-oscillator model of V5's known features (finite reach plus phase coupling), the answer is clean and depends on the reach in exactly the required way. When the reach is finite and smaller than the inter-group distance, chains within a group synchronize to a common rate while the two groups keep their different rates: local proper time, time dilation preserved, no universal clock. When the reach is made large enough to span both groups, everything locks and the between-group rate difference collapses to zero: a universal clock, which would destroy time dilation. V5's finiteness is what separates the physically required regime from the forbidden one.

Real-substrate confirmation. The phase model abstracts each chain to an oscillator. We then built a genuine cross-chain V5 term into the certified substrate simulator itself: chains on parallel lanes, each a real certified chain with the certified commitment-density update, their natural rates set by the carried-memory mechanism of Section 3 (more memory, slower advance), coupled by a term with V5's structure (finite transverse reach, finite longitudinal window, retarded). With a genuine within-group spread of natural rates (chains that would drift apart if uncoupled), the full signature reproduces from real substrate dynamics. When the reach is well below the group separation, the within-group rate spread collapses by 99 percent (the differently-paced chains lock to a shared rate, a real local proper time) while the between-group rate difference is preserved (time dilation intact). When the reach reaches the separation, the between-group difference collapses to near zero (a universal clock).

What this establishes, and what it does not. It establishes that V5's known structure, implemented as a real term in the certified substrate, produces local proper time without a universal tick, and that the reach being finite is load-bearing for time dilation. It does not derive from the primitives that the substrate must carry this term. The V5 coupling is an honestly-named addition, faithful to Paper_090's form but not forced by the 13 primitives. It also builds in the attractive (synchronizing) sign and uses relative position as a proxy for the gauge phase; both are flagged. Tier: STRUCTURAL, reverse-characterized (what V5 must be to yield the required physics), confirmed in the certified sim. The forward primitive derivation of V5 remains the hard open core (Section 7).

A necessity condition on V5. Working backward from the required physics gives a necessity condition, not a forward derivation: for V5 to bind chains into shared local clocks at all, its phase coupling must be attractive. A repulsive coupling gives no binding, and indeed the substrate's native, non-attractive coupling (Section 5) demonstrably anti-aligns rather than binds. Combined with the corpus's existing statement that

V5's reach is finite, this says V5 must be a finite-reach, attractive, retarded cross-chain coupling to yield the required physics. The attractive sign is assumed in the model here; its necessity is argued from that dispersive-coupling negative, not derived from the primitives. Tier: STRUCTURAL, a reverse-characterized necessity condition, not a derivation.

2. Why local clocks are needed: ED forbids universal becoming (prior result, consolidated)

Section 1's binder builds local clocks only because a universal one is forbidden. That prohibition is already derived in the ED gravity sector; we consolidate and cite it here.

The distinction that matters. ED makes commitment (P11) primitive and irreversible. P11 is the act of becoming, the event that turns an indeterminate degree of freedom into a determinate fact, ED's analogue of decoherence or measurement. What might be taken for a universal clock, P13, is not one: canonical P13 is time homogeneity, the statement that the substrate laws are invariant under time translation with no privileged moment. That is a symmetry, not a rhythm or a rate, and its stated role is to ensure that P11's commitment-direction is the only temporal asymmetry. ED does carry a fundamental time grain: P13's own second clause is a primitive event-discreteness (the substrate is not infinitely refinable), scaled by the substrate scale (P08). But a grain is a resolution, not a synchronized global clock that all chains obey in lockstep. There is no universal tick. Physics runs on where and when P11 fires, and the prohibition is against universal becoming: dense commitment in which P11 fires at every fundamental step. The corpus excludes that dense branch decisively.

The exclusion has three faces, stated in the ED gravity sector as a single structure (`foundations/Phase3_GR_SparseCommitme` `Phase3_GR_CommitmentIsSparse.md`).

Face 1: quantum mechanics, via the Zeno limit. If P11 fired at every tick, every degree of freedom would be measured every Planck time. That is exactly the quantum Zeno limit: continuous measurement does not merely weaken interference, it freezes evolution entirely. Dense commitment gives no persistent superposition, no interference, no coherence, no unitary evolution, and via Zeno no dynamics at all. Since ED has a quantum sector (P09 supplies superposition and the U(1) phase), the dense branch is not disfavored but incompatible with ED's own existence. Sparse commitment is forced, on pain of destroying ED's quantum mechanics. This is presented in the corpus as a rigorous consistency forcing, not a fit.

Face 2: gravitational time dilation. Becoming proceeds in proper time, and proper time runs at a position-dependent rate, the metric lapse. Gravitational time dilation just is becoming proceeding at different rates in different places. A uniform, dense, every-tick commitment would tick the same everywhere: no lapse variation, no dilation. It is the non-uniformity of becoming, the very thing that makes commitment sparse, that makes proper time and hence time dilation possible. (This observation, that it is hard to have time dilation if it ticks the same, is credited to Allen Proxmire in the corpus.)

Face 3: no preferred frame. The preferred-frame parameter α_1, the standard post-Newtonian quantity measuring how much a preferred or universal frame affects local gravity, is driven small by sparse commitment: α_1 is proportional to minus four times the commitment sparsity, of order 10^{-93} in the Solar System, safe by many tens of orders below the observational bound (see the GR-IV preferred-frame result; the value is a mechanism estimate, not a certified number). Sparse becoming makes the reference-frame number super small: with no detectable preferred-frame coupling, there is no universal frame.

Three corrections, recent self-fixes in the gravity sector, must be respected. First, the companion parameter α_2 is zero identically, but for a different reason: the luminal-cone condition (substrate signal speed equal to c), not sparse commitment. Only α_1 is the sparse-commitment number. Second, the preferred frame being spontaneously the cosmic microwave background rest frame fixes which frame is preferred; it does not make the couplings small. The smallness comes from sparse commitment (α_1) and the luminal-cone condition (α_2). Third, α_1's value is derived in mechanism but order-of-magnitude in value (the vacuum event-density magnitude is robust but unpinned, an order-one commitment factor is inherited, and no direct lunar-laser or pulsar test has been run); it is not a certified first-principles number.

The unified prior statement. Unitary becoming punctuated by rare irreversible commitments is one structure with three consequences: quantum mechanics, gravitational time dilation, and preferred-frame safety. ED forbids universal becoming, and this single prohibition is load-bearing across quantum mechanics, relativity, and dynamics. Sparse becoming is what keeps all three alive. This is prior work. What this paper adds is the constructive half: the mechanism, V5, that builds local clocks inside that world (Section 1), and an account of what mass is within it (Section 3).

The khronon is not a universal clock. ED's gravity sector carries a khronon, a preferred-time-foliation scalar. This is not an imposed absolute time. The khronon defines a local proper-time direction from its own local gradient; its background is spontaneously the cosmic rest frame, there being no other candidate, and finite-speed causation (V1) forbids a global now. Time dilation, the lapse, is full-strength ordinary metric gravity, and it is distinct from the suppressed khronon stiffness coupling that sources α_1. The foliation bends at full strength, so time dilation exists; only its stiffness coupling is suppressed by sparseness. The khronon is thus itself an instance of a local clock without a universal one, at cosmological scale.

3. What binds: mass as carried memory

Section 1 needs chains with natural rates of becoming that differ, and a notion of a chain's own tick that V5 can bind. Both come from a single mechanism, which also gives ED an account of mass it did not previously have.

The gap. The certified reference substrate is ballistic-or-extinct: a front either advances one step or extinguishes, with no slowed-but-surviving mode. This is why the two live substrate-Higgs candidates both failed on the reference substrate: there was no dispersion mode for a mass to inhabit. The diagnosis: the certified substrate has no chain-specific state that is both carried with the chain and read by its own dynamics. It has commitment density, which the dynamics read but which carries no chain identity, and it has an orientation, which travels with the chain but which the update rule is hard-coded to ignore. Neither can serve as a chain's memory of its own history.

The addition. Add exactly that missing channel: a chain-carried history value that the movement rule reads, updated with a finite memory (each commitment adds to it, and it fades). With this, a chain acquires a stable, bounded, surviving sub-ballistic rate: a real intrinsic-inertia signature, present with no external reference, intrinsic to the chain as mass should be. Higher memory yields more dwelling and a slower rate; the rate is the chain's own tick. Tier: CANDIDATE, positive, confirmed over long runs. It requires the added memory channel; it is not present in the bare substrate, and it is not the Standard Model Higgs sector.

The parameter, and where it bottoms out. The mechanism has a coupling strength and a fade rate. These are not independent: only their product is constrained, so there is effectively one free quantity, not two. The fade rate, moreover, is plausibly not a new unknown. V1's kernel is a chain's response to its own earlier state, so the memory fade is a candidate identification with V1's already-inherited memory time, here attached to working code for the first time (candidate, not proven, in the same sense as the mass-gravity identification below). The remaining free quantity reduces, dimensionally, to a single order-one factor. This is the same factor that recurs across the gravity preferred-frame coupling, quantum-computing decoherence rates, and interaction-vertex rates: an order-one commitment factor that the corpus inherits everywhere and derives nowhere. A dimensional check confirms that the three inherited substrate constants (signal speed, grain length, and action quantum) cannot form a pure number by themselves; a fourth ingredient, a rate, is required, and the natural candidate, the commitment rate, is itself the grain-scale rate times that same order-one factor. Tier: form FORCED, value INHERITED, with a real structural narrowing to a single recurring unknown.

Mass and gravity as two faces of one unknown. For a freely propagating chain the relevant commitment rate is the sparse one, and its suppression factor is the same quantity the gravity preferred-frame argument leaves inherited. Mass-memory fade and the gravitational sparse-commitment parameter plausibly share one underived number: the same order-one commitment factor and the same density scaling appear in both. This is a candidate identification, not proven. On the gravity side the relevant distinction is between always-on participation (P02) and sparse commitment (P11), two distinct canonical primitives contributing to the single scalar bandwidth of P04. That distinction was checked and found to rest on those canonical primitives, not on the archived four-band partition. (Canonical P04 is a single non-negative additive scalar with no bands; any four-band vocabulary traces to a retired M-series paper and is avoided here.)

4. Persistence is rhythmic

A chain persists not by sitting still but by continually re-committing. Re-committing with finite memory is a rhythm. This gives the chain's tick, the object V5 binds, a concrete form.

A single chain cannot oscillate in space: irreversibility forbids a front from reversing. The testable content is therefore temporal. With the carried-memory channel of Section 3, a chain's commitment pattern becomes a quantized, tunable, noise-robust stride: it advances on a sharp integer period, set by its memory strength, and that rhythm survives stochastic noise up to a large fraction of the signal scale before a sharp order-to-disorder transition. To persist is to pulse: a persisting chain does not drift smoothly, it ticks, at a memory-set rate. Tier: MEASURED, confirmed in the substrate sim, conditional on the added memory channel. Ceiling: this is a temporal rhythm, a rate of becoming, not a spatial wave and not full phase-wave (de Broglie) behavior, since the phase channel is not read by the certified dynamics.

5. Why not global: the collective negative is correct physics

If chains tick, and V5 can bind nearby ones, one might ask whether many chains synchronize into a single global pulse. They do not, and that is the physically required answer.

Under the substrate's own native coupling (shared commitment density plus excluded volume), independent chain-rhythms do not globally synchronize. The coupling is dispersive, traffic-like: nearby chains hitting each other's dense trails or blocking each other spread their rates apart rather than pulling them together. A phase-randomized surrogate test confirms there is no genuine collective synchronization; the aggregate periodicity is fully explained by the chains being individually rhythmic, and the coupling if anything mildly de-aligns them. Separately, on a closed system the irreversible growth of commitment density saturates the medium and eventually stops even the single-chain rhythm.

This negative is exactly what Section 2 requires. Global synchronization would be universal uniform becoming, which is forbidden (Zeno freeze, loss of time dilation, loss of dynamics). The substrate correctly refuses a universal clock. The right question was never global synchronization but the relational one: local binding without universal becoming, which V5's finite reach supplies (Section 1). The native dispersive coupling cannot do it; a genuine finite-reach V5 coupling can.

6. Synthesis: mass, time, and sparse becoming are one story

Physics requires sparse, local, relational becoming, never universal uniform becoming. The gravity sector already unifies three consequences of that constraint: quantum mechanics (avoiding the Zeno freeze), time dilation (position-dependent rate of becoming), and no preferred frame (the preferred-frame parameter suppressed to roughly 10^{-93}). This paper adds a fourth face and the constructive mechanism behind all of them at the local scale. A chain's own carried memory gives it inertia and a proper-time rhythm: that is mass. V5's finite reach binds nearby chains into a shared local rate: that is a composite rest frame, and the shared local clock whose position-dependence is time dilation. V5's finiteness prevents that shared clock from ever becoming universal.

This unification is structural coherence, not confirmation. Both the carried-memory channel and the V5 coupling are honest additions to the certified substrate, not features derived from the 13 primitives. Reproducing the target behavior with them demonstrates that the pieces fit consistently; it is not evidence that the substrate carries them. In particular, V5 was reverse-characterized to produce local proper time, so its producing local proper time is coherence, not an independent confirmation. The load-bearing open question (Section 7) is precisely whether the primitives force these additions.

With that caveat stated plainly: the two halves fit. Sparse becoming forbids a universal frame; V5's finite reach builds the local ones instead. Mass, and the composite rest frames that carry time dilation, are the same phenomenon seen from V5, chains acquiring a shared local rate of becoming.

This narrows the program's highest-leverage open target, the characterization of V5, from uncharacterized to a precise remaining ask. V5 must be a finite-reach, attractive, retarded cross-chain coupling that binds local proper time while preserving time dilation. The relational and dynamical characterization is done, in the certified substrate. What remains is not a forward derivation of V5's existence: V5 is a kernel rule-type under P10, posited exactly as V1 is (Paper_089), so the primitives no more force V5 to exist than they force V1. What remains open sits upstream, in the operational definition of P12's coherence term, on which the status of V5's attractive sign turns (Section 7).

7. Open frontier

- V5's existence, stated precisely: V5 is a **kernel rule-type under P10**, whose existence is *posited*, exactly as V1's is (Paper_089). P10 admits cross-chain kernel rule-types but forces no particular one, so "do the primitives force V5 to exist" has the same answer as the same question about V1: no, by the nature of P10. What is forced is V5's *form* (finite-reach from P05 locality, retarded from P11, gauge-covariant relative phase from P05 + P09, bounded from P04); the attractive sign stands as a *necessity condition* (no binding without it), not a forward derivation of the sign.
- The upstream fork the sign turns on: **P12's coherence term (Coh) has no operational definition anywhere in the corpus.** Its canonical gloss is "alignment / phase-consistency content," which would host V5's attractive coupling as the cross-chain face of Coh; but its only operational instance, the certified simulator, reads commitment density only and is polarity-blind by hard invariant. If Coh is genuinely phase-based, V5's sign becomes derivable, at the cost of making Σ read flow-direction and revising the orientation-blindness invariant; if Coh is density-based, V5 is permanently a P10 addition. That fork, not V5 itself, is the open core.

Update 2026-07-08 (attractive sign supplied; fork resolved, and more cheaply than stated above). The fork is resolved, favorably, and the anticipated "cost" turned out not to apply. The standalone result *Phase Coherence in ED: Attractive Sign, Finite Reach, and Substrate Disorder* (`physics-papers/substrate-evaluation/Paper_PhaseCoherence_P12Coh.md`) build-verifies, on the certified Σ-rule in 2D and 3D, that P12's coherence term is phase-based and alignment-rewarding, **without revising the orientation-blindness invariant**: that invariant governs the substrate's spatial *director* (B5), not P09 polarity (the amplitude phase π_K), which is a distinct, weakly-associated channel the single-rule-type simulator does not represent. The alignment-rewarding coherence term supplies V5's attractive/synchronizing sign, and it stays finite-reach (Knots-safe, which is exactly V5's local-not-universal binding) because the P05 connection carries the substrate's own disorder and commitment is irreversible. So the attractive sign moves from a *necessity condition* to a *build-verified consequence* of the P12-Coh operationalization (MEASURED tier; form-forced conditional on the operationalization; reach value-inherited). What remains open is only V5's *existence* as a P10 kernel-rule-type posit, unchanged. - Whether the memory channel and the V5 coupling can be reduced to existing primitives rather than added. Both are honest additions here. - The single order-one commitment factor: derive it once, and it fixes the mass-memory scale, the gravity preferred-frame coupling, decoherence rates, and vertex rates together. - Full phase-wave (de Broglie) behavior, which requires the phase channel that the certified dynamics do not read; the rhythm established here is temporal only. - The candidate identity between the mass-memory fade rate and the gravitational sparse-commitment parameter: same participation-vs-commitment distinction, same order-one factor, same density scaling, not yet proven a single scalar. - Collective pulsation in an open, throughput system. A star is open; the closed substrate saturates. Whether local rhythms lock into a sustained collective pulse in an open system is untested.

Load-bearing step audit

#	Step	Tier	Basis
1	Sparse becoming forced: dense commitment is the Zeno freeze, no QM; the three faces (QM, time dilation, no preferred frame)	DERIVED (prior corpus, cited)	Phase-3 GR sparse-commitment files
1a	The α_1 magnitude ($\approx 10^{-93}$) that quantifies preferred-frame safety	MECHANISM ESTIMATE, value uncertified (order-of-magnitude, untested)	Section 2, Face 3; GR-IV
2	Khronon is a local foliation, spontaneously the cosmic frame, not an imposed universal clock; lapse (time dilation) is distinct from the suppressed α_1 coupling	STRUCTURAL (prior corpus, cited)	GR khronon / MOND-reconciliation notes
3	Certified substrate is ballistic-or-extinct, no dispersion mode	MEASURED	E1 and H1-leg results
4	Chain-carried, dynamics-read memory gives a bounded surviving sub-ballistic rate (mass)	CANDIDATE (requires the added, non-primitive memory channel)	Section 3
5	Memory parameters reduce to one product; fade rate is a candidate identification with V1's inherited memory time; bottoms out on one order-one factor	STRUCTURAL and INHERITED (with a candidate identification)	Section 3
6	Finite memory gives a quantized, noise-robust temporal rhythm	MEASURED (in sim, conditional on the added channel)	Section 4
7	Native coupling does not globally synchronize (correct: no universal becoming)	MEASURED and STRUCTURAL	Section 5
8	V5's finite reach gives local proper time and preserved time dilation, in the certified substrate	STRUCTURAL, reverse-characterized, confirmed in sim	Section 1
9	V5 must be finite-reach and attractive (necessity conditions given the target physics)	STRUCTURAL (reverse-characterized necessity, not derivation)	Section 1

#	Step	Tier	Basis
10	V5's existence: a P10 kernel-rule-type posit (like V1); attractive sign stands as a necessity condition; operationalizing P12-Coh is the upstream fork	POSITED (existence) / NECESSITY (sign) / OPEN (Coh)	Section 7

Methods and reproducibility

All substrate probes reuse the certified simulator's per-decision functions (candidate enumeration, the Σ functional, the deterministic tie-break, and the commit chokepoint) verbatim; only the memory channel and the V5 coupling are added, and both are flagged as additions throughout. Probe scripts and full results files are listed below.

- **Mass / carried memory** (`dwell_channel_mass_probe.py`, `dwell_intrinsic_memory_probe.py`; results `Dwell_Intrinsic_Memory_Results.md`): single certified chain of 400+ nodes, self-loop channel, memory update new = decay × old + increment; effective velocity averaged over the last 1000 of 8000 steps, 6 seeds; the plateau $v_{\text{eff}} = 0.72 \pm 0.02$ across seeds.
- **Rhythm** (`dwell_memory_rhythm_probe.py`; `Dwell_Memory_Rhythm_Results.md`): chain of 20000 nodes, 15000 steps, first 3000 discarded, 6 seeds; inter-hop-interval coefficient of variation as the clockwork-vs-random estimator; noise robustness swept by tie-break jitter from 10^{-3} to 1.0 (order-disorder transition between 0.3 and 1.0× the signal scale).
- **Collective (native coupling)** (`collective_pulse_probe.py`; `Collective_Pulse_Results.md`): ring of 3000 nodes, fresh pre-saturation window [200,1000] verified by a mean-ρ tracker, 5 seeds; phase-randomized surrogate (100 surrogates per seed, each front's rhythm preserved and only its phase offset randomized) as the synchronization null; real collective oscillation amplitude $z = -1.0$ to -1.8 versus surrogate (no synchronization).
- **V5 coupling** (`v5_synchronization_probe.py` phase model; `v5_substrate_coupling_probe.py` real substrate; `V5_Synchronization_Characterization.md`): real-substrate run is 8 parallel certified lanes (4000 nodes each), two clusters of 4 at transverse separation 20, genuine within-cluster natural-rate spread (memory values giving quantized rates $\sim 0.67/0.5/0.33$), 3 seeds, 2500 steps, transverse reach ℓ_{V5} swept 0.5 to 100. Within-cluster rate spread collapses from 0.084 (uncoupled) to 0.0005 (mean over seeds) at finite reach below the separation, while the cross-cluster rate gap holds at ~ 0.25; the gap collapses to ~ 0.002 once reach reaches the separation. The 99 percent within-cluster collapse is the seed-averaged figure.

Estimators, seeds, and grids are stated so each result is rerunnable from the cited scripts.

Reference example for house template and tone: `Paper_CommonCauseNotChannel_A1.md`. Build the PDF with the standard pandoc plus xelatex recipe when the draft is finalized.